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Mechanistic *In-Silico* Insights into the Anti-quorum Sensing Potential of Coumaric Acid and Syringic Acid in *Serratia marcescens* with Invitro Analysis

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Abstract

Antibiotic resistance presents a major global health threat, especially with ESKAPE pathogens like *Serratia marcescens*, which exhibit resistance to all known antibiotics. Quorum sensing (QS) is key to its virulence and resistance, emphasizing the need for novel natural antimicrobial agents. This study investigates two plant-derived phenolic compounds, coumaric acid and syringic acid, targeting QS proteins of *S. marcescens* using *in silico* molecular docking, molecular dynamics simulations, and in vitro biochemical assays. Validated homology models of eight QS-associated proteins—BsmA, FimA, FimC, FlhD, LuxS, PigP, RsmA, and RssB—were employed for molecular docking studies, ADME (absorption, distribution, metabolism, and excretion) profiling, and 100-ns molecular dynamics (MD) simulations to evaluate ligand-binding stability. Coumaric acid displayed more desirable physicochemical properties (logP 1.79; TPSA 57.53 $Å^2$) compared to syringic acid (logP 1.04; TPSA 75.99 $Å^2$). Binding energy calculations indicated a stronger affinity of coumaric acid for six of the proteins, with the LuxS–coumaric acid complex showing the most significant interaction (ΔGbind -21.74 ± 3.01 kcal/mol). Analysis of the MD trajectory revealed that coumaric acid enhanced protein stability, as shown by reductions in RMSF (root mean square fluctuation), a more compact Rg (radius of gyration), decreased SASA (solvent-accessible surface area), alterations in the Dictionary of secondary structure of protein (DSSP), and consistent hydrogen bonding. Conversely, syringic acid induced increased conformational flexibility and destabilized alpha-helices and beta-sheets in specific proteins. Principal component analysis (PCA) confirmed tighter conformational clustering in coumaric acid complexes, consistent with improved stabilization. Furthermore, antibacterial assays demonstrated strong inhibitory effects, with MIC values of 700 µg/mL for coumaric acid and 1000 µg/mL for syringic acid. Coumaric acid displayed a bactericidal effect, whereas syringic acid was bacteriostatic. Additionally, time–kill assays revealed a dose-dependent reduction in bacterial growth over 48 h following treatment with varying concentrations of these phenolic acids . Interestingly, qPCR analysis of QS-specific gene expression showed significant downregulation of key QS-regulated genes in response to both compounds, highlighting their potential as quorum-sensing inhibitors and supporting their development as alternative antimicrobial agents against antibiotic-resistant *S. marcescens*.

Keywords Coumaric acid · Syringic acid · Quorum sensing · Antibiotic resistance · Molecular dynamic simulations · Antiquorum sensing agents

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1 Introduction

The advent of antibiotics marked a transformative milestone in modern medicine; however, the escalating threat of antimicrobial resistance (AMR) now poses a formidable global health crisis. The emergence and spread of multidrug-resistant (MDR) bacteria have rendered many standard treatments ineffective, leading to prolonged illness, increased healthcare costs, and higher mortality, particularly in vulnerable populations and low-resource settings [1–6]. Antibiotic resistance has emerged as a critical global health concern, driven by the widespread misuse and overuse of antimicrobials across clinical and agricultural settings. The World Health Organization estimates that AMR is responsible for 1.27 million deaths annually, with projections suggesting up to 10 million deaths per year by 2050 if unaddressed [7, 8]. AMR arises through diverse mechanisms, including genetic mutations, horizontal gene transfer, and adaptive responses to the selective pressure exerted by the widespread misuse of antibiotics [9]. A significant contributor to this crisis is the rise of ESKAPE pathogens—*Enterococcus faecium*, *Staphylococcus aureus*, *Klebsiella pneumoniae*, *Acinetobacter baumannii*, *Pseudomonas aeruginosa*, and *Enterobacter spp*.—which are known for their ability to "escape" the effects of multiple antibiotics and cause the majority of hospital-acquired infections. Among these, *Serratia marcescens* has drawn increasing concern due to its intrinsic resistance mechanisms, robust biofilm formation, and growing multidrug resistance, significantly complicating the treatment of nosocomial infections such as pneumonia, bloodstream, and urinary tract infections [10].

Serratia marcescens, a Gram-negative opportunistic pathogen of *Enterobacteriaceae* family, has become an increasingly problematic member of the ESKAPE group due to its growing antibiotic resistance and prevalence in hospital-acquired infections. Globally, resistance rates in *S. marcescens* have escalated, with surveillance reports indicating that over 40 %–60 % of clinical isolates are resistant to third-generation cephalosporins, aminoglycosides, and even carbapenems in some regions. Its resistance mechanisms include the production of β-lactamases (such as AmpC), efflux pumps, and reduced membrane permeability, enabling survival against multiple antibiotic classes [11].

Serratia marcescens employs quorum sensing (QS) as a critical regulatory system to coordinate gene expression in a cell-density-dependent manner, playing a pivotal role in its virulence, biofilm formation, and antibiotic resistance. The primary QS system in *S. marcescens* is the SmaIR system, homologous to the LuxIR system, where SmaI synthesizes N-acyl homoserine lactones (AHLs), and SmaR acts as the receptor/regulator that controls the expression of target genes [12].

This QS system governs the production of several virulence factors, including proteases, hemolysins, and prodigiosin, and critically modulates biofilm formation—a key contributor to antibiotic resistance. Within biofilms, cells are embedded in a self-produced extracellular matrix that hinders antibiotic penetration and facilitates persistence under antimicrobial stress. Moreover, QS enhances the expression of efflux pump genes and β-lactamases, which further promotes resistance to antibiotics such as β-lactams, aminoglycosides, and fluoroquinolones. Targeting quorum sensing (QS) in *Serratia marcescens*, particularly the SmaIR system, offers a promising strategy to disrupt biofilm formation and virulence without promoting resistance, by blocking AHL signal synthesis or receptor binding [13–15]. This highlights the urgent need for novel, plant-derived antimicrobials that can interfere with QS pathways and serve as effective, resistance-free therapeutic alternatives.

Plant derived phenolic acids such as coumaric acid and syringic acid have been identified as potent QS inhibitors due to their capacity to modulate bacterial cell-to-cell communication. These phytochemicals disrupt N-acyl homoserine lactone (AHL)-mediated signalling pathways, inhibit biofilm maturation, and downregulate QS-controlled virulence gene expression in multiple Gram-negative bacteria. Although their molecular mechanisms in *Serratia marcescens* QS regulation remain insufficiently characterized, these compounds are postulated to interfere with QS signal transduction cascades and transcriptional regulators, thereby attenuating pathogenic phenotypes and biofilm-associated persistence. Given their multi-targeted bioactivity and low cytotoxicity, coumaric acid and syringic acid represent promising candidates for the development of novel anti-virulence therapeutics aimed at mitigating antimicrobial resistance [16–24].

In this current study, an integrative *in silico* and in vitro approach was employed to elucidate the QS inhibitory potential of coumaric acid and syringic acid against *S. marcescens*. Molecular docking studies quantified binding affinities and interaction profiles of these compounds with eight critical QS-related proteins—bsmA, fimA, fimC, flhD, LuxS, PigP, rsmA, and rssB—followed by molecular dynamics simulations to assess the conformational stability and dynamic behaviour of protein–ligand complexes under physiologically relevant conditions. These computational insights were substantiated by in vitro antibacterial and anti-QS assays, to validate the efficacy of these phenolics in modulating QS-regulated bacterial functions and providing a potential strategy to combat *S. marcescens*-associated infections.

2 Materials and Methods

2.1 Protein Modelling

To investigate the molecular interactions of eight key QS-related proteins with coumaric and syringic acid, their three-dimensional structures were first queried in the Protein Data Bank (https://www.rcsb.org/). As no experiment ally resolved structures were available, homology modelling was carried out. The FASTA sequences of the target proteins—bsmA (102 aa, UniProt ID: A0A379Y3G6), fimA (180 aa, P22595), fimC (242 aa, A2TEG2), flhD (119 aa, O85806), LuxS (171 aa, Q684Q1), PigP (204 aa, C1IZ16), rsmA (272 aa, A0A656VQ38), and rssB (219 aa, Q8GP20)—were retrieved from UniProt. Structural prediction and functional annotation were subsequently carried out using the I-TASSER web server [25, 26]. Following structure prediction with I-TASSER, the generated models provided secondary structure information along with confidence (C) scores. Higher C scores indicate more reliable predictions of the protein structure (Table 1). Additionally, to ensure structure validity, the stereochemical quality of the modelled proteins was evaluated using Ramachandran plots generated through the MolProbity web server [27].

2.2 MD Simulation of Naked Protein

Molecular dynamics (MD) simulations of the modelled protein structures were performed for 100 ns using GROMACS 2023.2 [28]. The CHARMM36 force field and the TIP3P water model were applied in a triclinic box, with counter ions added to neutralise system charges. Energy minimization was carried out using the steepest descent algorithm over 50,000 steps, followed by equilibration under NVT and NPT ensembles (100 ps each; 50,000 steps) at 300 K and 1 bar with a 2 fs time steps to keep the number of particles (N), volume (V), and temperature (T) constant as per canonical and isothermal-isobaric conditions [29]. MD simulation of the naked protein was served as a reference for comparison with ligand-bound complexes. The energy-minimized protein structures were subsequently employed for docking studies with coumaric acid (PubChem ID: 637542) and syringic acid (PubChem ID: 10742).

Table 1 C score of the predicted protein structure by I-TASSER

Protein	C-Score
bsmA	−2.13
fimA	−0.53
fimC	−0.48
flhD	−0.02
luxS	1.43
pigP	−0.33
rsmA	1.06
rssB	1.44

2.3 Preparation of Ligand

The structure data files (sdf) of ligand molecules; coumaric acid & syringic acid, were downloaded from PubChem and energy minimized using UCSF-Chimera [30, 31]. In silico prediction of Absorption, Distribution, Metabolism, and Excretion (ADME) properties was performed using the SWISS ADME web server (https://www.swissadme.ch/). ADME profiling represents a critical aspect of pharmaceutical research and development, as a comprehensive understanding of absorption, distribution, metabolism, and elimination is essential for evaluating a compound's therapeutic efficacy and safety. These parameters are essential for evaluating drug bioavailability, therapeutic efficacy, safety, and potential drug–drug interactions [32].

2.4 Docking

The energy-minimized naked protein structures were used for molecular docking. The binding pocket coordinates were first predicted using C-Dock2, which performs blind docking and ranks the top five potential binding pockets for each protein based on Vina scores [33]. The coordinates of the predicted binding site with the lowest Vina score was selected, and its coordinates were used for docking ligands onto the protein structures using the AutoDock Vina plugin implemented in UCSF Chimera [34].

The protein structures were prepared for docking using the Dock Prep tool in UCSF Chimera, which effectively substituted any missing side chains by employing the Dunbrack 2010 rotamer library [35]. The processed proteins were saved in. mol2 file format. Ligands were similarly prepared by energy minimization and hydrogen atoms addition before being saved in. mol2 format. Both proteins and ligands were subsequently opened in UCSF Chimera, and the site-specific docking was performed using the AutoDock Vina plugin. The binding pocket coordinates having lowest vina score predicted by CB- Dock2 were entered, and a grid box was generated around the site to define the ligand-binding region. AutoDock Vina employs a gradient-based conformational search algorithm with scoring derived from the quasi-Newton Broyden-Fletcher-Goldfarb-Shanno (BFGS) method and the X-score algorithm [36, 37]. Docking results comprised binding energies and five predicted ligand poses. The docked pose with the lowest binding energy was saved as a.pdb file and subsequently used for molecular dynamics (MD) simulations. The docked complexes were further refined using Maestro (Schrodinger®), which merged docked ligand and protein poses into a single pdb file, and a two-dimensional interaction map was generated with BIOVIA® Discovery studio [38]. Furthermore, BIOVIA® Discovery studio was employed to visualize the hydrogen donors and acceptors, as well as ligand–protein complex solvent accessibility [38]. To further characterize binding interactions, the Protein–Ligand Interaction Profiler (PLIP) was used to identify key hydrogen bonds, hydrophobic interactions, and the specific residues involved [39].

2.5 Molecular Dynamic Simulation of Protein–Ligand Complex

MD simulations were performed to investigate the interactions of coumaric acid and syringic acid with eight target proteins: bsmA, fimA, fimC, flhD, luxS, pigP, rsmA, and rssB. All simulations were carried out using GROMACS 2023.2 [40] with the CHARMM36 force field, which is widely recognized for its accuracate modelling of biological macromolecules [41].

Each protein–ligand complex was embedded in a triclinic simulation box solvated with TIP3P water molecules to create a biologically realistic environment. Counter ions were added to neutralize the overall system charge. Energy minimization was then performed using the steepest descent method for 50,000 steps, reducing steric clashes and unfavourable contacts and bringing the maximum force below 1000 kJ/mol/nm therby yielding a stable starting conformation for subsequent simulations.

System equilibration was performed in two phases. First, NVT equilibration was conducted to maintain constant particle number, volume, and temperature, followed by NPT equilibration to allow the system to adjust to reach constant pressure. Each equilibration phase was run for 100 picoseconds, with a 2 femtosecond timestep. Temperature was maintained at 300 K using a velocity-rescaling thermostat, and pressure was controlled at 1 bar with Parrinello-Rahman barostat, ensuring proper system stabilization before initiating the production runs [42, 43]. For the production phase, the same velocity-rescaling thermostat and Parrinello-Rahman barostat, were applied with relaxation times to 0.1 ps for temperature and 2.0 ps for pressure, providing stable and accurate sampling conditions.

To preserve molecular geometry during simulations, all bond lengths involving hydrogen atoms were constrained using the LINear Constraint Solver (LINCS) algorithm. Non-bonded interactions were calculated using the Verlet cutoff scheme (12 Å short-range cutoff), while long-range electrostatic interactions were treated using the Particle Mesh Ewald (PME) method.

Periodic Boundary Conditions (PBC) were applied in all three dimensions to minimize artifacts from system edges. Production simulations were run for 100 ns, with trajectories and energy data saved every 10 picoseconds for downstream analysis [44].

Data visualisation was carried out using Grace software (available at https://plasma-gate.weizmann.ac.il/Grace) and protein–ligand binding free energies were estimated using the gmx_MMPBSA tool [45].

Binding energy gives insight into the strength and stability of the protein–ligand interaction. It was calculated using the following equation:

$$\Delta G_{bind} = \langle G_{COM} \rangle - \langle G_{REC} \rangle - \langle G_{LIG} \rangle \tag{1}$$

here ΔG_{bi} is the binding free energy, where COM refers to the complex (protein + ligand), REC to the receptor (protein), and LIG to the ligand alone.

Each energy term is derived from:

$$\langle G_x \rangle = \langle E_{MM} \rangle + \langle G_{sol} \rangle - \langle TS \rangle \tag{2}$$

where x can be COM, REC, or LIG. EMM represents the molecular mechanics energy, $Gsol$ is the solvation free energy, T is the absolute temperature, and S is the entropy.

This expression can be simplified to the thermodynamic relationship:

$$\Delta G_{bind} = \Delta H - T\Delta S \tag{3}$$

where ΔH is the binding enthalpy, and $T\Delta S$ accounts for the entropic contribution. For comparative purposes, especially between similar ligands, it is common to focus on ΔH and omit the entropy term when approximating relative binding strengths.

The enthalpy component can be further decomposed as:

$$\Delta H = \Delta E_{MM} + \Delta G_{sol} \tag{4}$$

with:

$$\begin{aligned} \Delta E_{MM} &= \Delta E_{bonded} + \Delta E_{nonbonded} \\ &= (\Delta E_{bond} + \Delta E_{angle} + \Delta E_{dihedral}) \\ &\quad + (\Delta E_{electrostatic} + \Delta E_{\text{van der waal}}) \end{aligned} \tag{5}$$

The solvation energy consists of both polar and non-polar components:

$$\begin{aligned} \Delta G_{sol} &= \Delta G_{polar} + \Delta G_{non-polar} \\ &= \Delta G_{GB} + \Delta G_{non-polar} \end{aligned} \tag{6}$$

where GB stands for the Generalized Born model used for estimating polar solvation effects.

The non-polar solvation energy can be expressed in two ways:

Based on solvent-accessible surface area (SASA):

$$\Delta G_{non-polar} = NP_{TENSION} * \Delta SASA + NP_{OFFSET} \tag{7}$$

Or as a sum of dispersive and cavity terms:

$$\Delta G_{non-polar} = \Delta G_{dispersion} + \Delta G_{cavity} = \Delta G_{dispersion} + (CAVITY_{TENSION} * \Delta \mathrm{SASA} + CAVITY_{OFFSET}) \tag{8}$$

In these formulations, ΔE_{MM} captures both bonded and non-bonded interactions in the gas phase, while ΔGsol accounts for the solvent effects.

For MMGBSA calculations, the Generalized Born model was applied, with AMBER topologies generated from GROMACS files. The calculations used the following parameters: PBradii = 3, internal dielectric constant = 1.0, an external dielectric constant = 78.5, and surface tension of 0.0072. Protein and ligand force fields were assigned using oldff/leaprc.ff99SB and leaprc.gaff respectively. The energy data were visualized and analysed the results through the graphical interface of gmx_MMPBSA_ana [45]. To further examine dynamic behaviour and structural variation in the protein–ligand complexes, Principal Component Analysis (PCA) was performed using MODE-TASK software [46]. Secondary structure composition was analysed using the Dictionary of Secondary Structure of Proteins (DSSP) algorithm, which classifies amino acid residues based on their secondary structure assignments [47]. When integrated with GROMACS, DSSP facilitates the examination of protein structural dynamics during simulations, providing insights into the temporal evolution of secondary structure elements upon ligand interactions [48, 63].

2.6 Trajectory Analysis

Trajectory analysis was performed using a combination of specialized tools, including VMD, PyMOL, UCSF Chimera, UCSF ChimeraX, BIOVIA Discovery Studio, Schrödinger Maestro, Xmgrace, and GROMACS**.** Several structural and dynamic parameters were calculated to evaluate the stablility and behaviour of protein–ligand complexes during MD simulations. Root Mean Square Deviation (RMSD) was used to check the overall structural stability, while Root Mean Square Fluctuation (RMSF) quantified residue-level flexibility. The Radius of Gyration (Rg) was calculated to monitor structural compactness, and Solvent-Accessible Surface Area (SASA) was measured to evaluate solvent exposure. Hydrogen bond (HB) analysis was performed to quantify the number of HBs formed during the molecular dynamics (MD) simulations. The temporal evolution of secondary structures (α-helices, β-sheets, turns, and coils) following ligand binding was assessed using DSSP data extracted from GROMACS trajectories and analysed with a Python script (https://github.com/drbipinraj/DSSP-Plot/tree/main). Principal Component Analysis (PCA) was performed to examine large-protein motions and binding free energies were estimated using the Molecular Mechanics Generalized Born Surface Area (MMGBSA). These analyses provided a comprehensive understanding of the dynamic stability and interaction strength of the complexes.

2.7 Software and Tools

Protein Sequence retrieval: Uniprot website (www.uniprot.org), Protein structure modelling; I-TAESSAR (https://zhanggroup.org/I-TASSER/), Ramachandran Plot: MolProbity (http://molprobity.biochem.duke.edu/), Ligand structure retrieval: PubChem (https://pubchem.ncbi.nlm.nih.gov/), Ligand Binding Pocket prediction: CBDock2 (https://cadd.labshare.cn/cb-dock2/php/blinddock.php), ADME (https://www.swissadme.ch/), Python code to compute DSSP plot (https://github.com/drbipinraj/DSSP-Plot/tree/main), Molecular docking: UCSF-Chimera, AutoDock Vina.

Molecular dynamics simulation: GROMACS-2023.

PCA and RMSD graph- MODE-TASK.

Binding Free energy calculation: gmx_MMGBSA.

Molecular visualization: VMD, PyMOL, UCSF-Chimera, UCSF-ChimeraX, Biovia Discovery Studio, Schroodinger- Maestro.

Graph- Xmgrace, Graphpad Prism.

Antibacterial studies phenolic acids against S. marcescens: Coumaric acid and syringic acid were purchased from Sigma-Aldrich and used in their original form. Working stock solutions were freshly prepared as required from a primary stock solution at a concentration of 10 mg/mL.

Microbial species and their cultivation conditions: *Serratia marcescens* strain was obtained from the Microbial Type Culture Collection (MTCC) in Chandigarh, India having id MTCC2645. During all experimental procedures, the bacterium was cultured in Growth Medium 3 (GM3) liquid broth and maintained on 2% agar plates under controlled conditions at 37 °C, with continuous shaking at 120 rpm to promote optimal growth and maintain viability.

Minimum inhibitory concentration (MIC) and minimum bactericidal concentration (MBC) of phenolic acids against *S. marcescens*: To determine the minimum inhibitory concentration (MIC) of coumaric acid and syringic acid against *Serratia marcescens* was determined using a modified microdilution method in accordance with Clinical and Laboratory Standards Institute (CLSI) guidelines. A bacterial suspension (10^6 CFU/mL) was inoculated into 96-well plates containing serial dilutions of each compound (0–1000 µg/mL) and incubated at 37 °C with gentle shaking (100 rpm) for 24 h. Post-incubation, bacterial viability was evaluated using the MTT assay (3-(4,5-Dimethylthiazol-2-yl)-2,5-Diphenyltetrazolium Bromide), following established protocoldescribed in previous studies [49–52].

The minimum bactericidal concentration (MBC) was determined according to the standard CLSI (2019) guidelines. From the MIC test, 0.1 mL of the inoculum from wells showing no visible bacterial growth were spread onto plain GM3 agar plates and incubated at 37 °C for 24 h. The lowest concentration of coumaric acid or syringic acid that completely inhibited colony formation was recorded as the MBC [53–55].

Growth Kill Assay: A growth-kill assay was conducted to assess the temporal effects of coumaric acid and syringic acid on bacterial growth. In this study, bacterial cultures were serially diluted to a final concentration of 10^8 CFU/mL, after which they were exposed to varying concentrations of coumaric acid and syringic acid (ranging from 0 to 2000 µg/mL) in 96-well microtiter plates. The assay was performed under controlled conditions, and optical density (OD) readings at 595 nm were recorded at regular intervals over a 48-h period. Growth curves were generated by plotting absorbance values against the concentrations of the phenolic acids, providing insights into their dose-dependent inhibitory effects on bacterial proliferation [56].

qRT-PCR Analysis of Anti-Quorum Sensing Potential of Phenolic Acids in *S. marcescens*: Quantitative real-time PCR (qRT-PCR) was performed to assess the anti-quorum sensing activity of coumaric acid and syringic acid by evaluating their effects on the expression of genes related to biofilm formation, adhesion, motility, and virulence in *Serratia marcescens*. Transcript levels of luxS, pigP, rsmA, flhD, fimC, bsmA, and fimA were analysed using gene-specific primers (listed in Table 2). Bacterial cultures were incubated at 28 °C for 24 h, then harvested by centrifugation at 5000×*g* for 5 min at 25 °C. Total RNA was extracted from the cell pellets using the NucleoSpin RNA kit (Takara) and subsequently converted into cDNA using the Qiagen cDNA synthesis kit. The qRT-PCR was performed using Bio-Rad's real-time master mix, with rplU serving as the housekeeping gene for normalization. Relative changes in gene expression was calculated using the $2^{-\Delta\Delta Ct}$ method, where ΔCt is the difference in cycle threshold (Ct) between the target gene and rplU, and ΔΔCt compares these values between treated and untreated (control) samples [57].

2.8 Statistical Analysis

All assays were conducted with two biological replicates and three technical replicates. Data are presented as mean values. Statistical significance was determined using one-way analysis of variance (ANOVA), with thresholds set at $*P<0.05$, $**P<0.01$, and $***P<0.001$.

3 Results

3.1 Homology Modelling and Docking

The predicted 3D structures of all eight proteins (shown in Fig. 1) were first validated using Ramachandran plot analysis (Supplementary Fig. 1) to ensure their structural quality. After validation, the structures underwent energy minimization before evaluating their binding affinities with coumaric acid and syringic acid. ADME predictions revealed that coumaric acid possesses favourable pharmacokinetic properties, with a logP of 1.79, a TPSA of 57.53 Å^2, and a pKa of 4.64, indicating high absorption and bioavailability. In comparison, syringic acid showed higher predicted absorption but reduced bioavailability, as reflected by its logP of 1.04, TPSA of 75.99 Å^2, and pKa

Table 2 Primers used in the study of gene expression in *Serratia marcescens*

Gene	Product	Primer sequence
luxS	Autoinducer-2 production through quorum sensing	Forward: GCTGGAACACCTGTTCGC Reverse: ATGTAGAAACCGGTGCGG
fimA	Helps in adhesion	Forward: GGTGAAATTGTACAGTCCACCTG TA Reverse: TCACATTTCTCCAGGCTAATATCG
flhD	Regulate flagellar motility	Forward: TGTCGGGATGGGGAATATGG Reverse: CGATAGCTCTTGCAGTAAATGG
pigP	Helps in pigment production	Forward: GAACATGTTGGCAATGAAAA Reverse: ATGTAACCCAGGAATTGCAC
fimC	Fimbrial protein helps in attachment to surface	Forward: ACCAGCCGTTTCAACAACAA Reverse: GGTTTGTACGGTGCGATCTT
rsmA	Helps in regulation of swarming motility	Forward: TTGGTGAAACCCTCATGATT Reverse: GCTTCGGAATCAGTAAGTCG
bsmA	Regulate biofilm formation	Forward: CCGCCTGCAAGAAAGAACTT Reverse: AGAGATCGACGACGGTCAGTTCC
rpu1	Housekeeping gene	Forward: GCTTGGAAAAGCTGGACATC Reverse: TACGGTGGTGTTTACGACGA

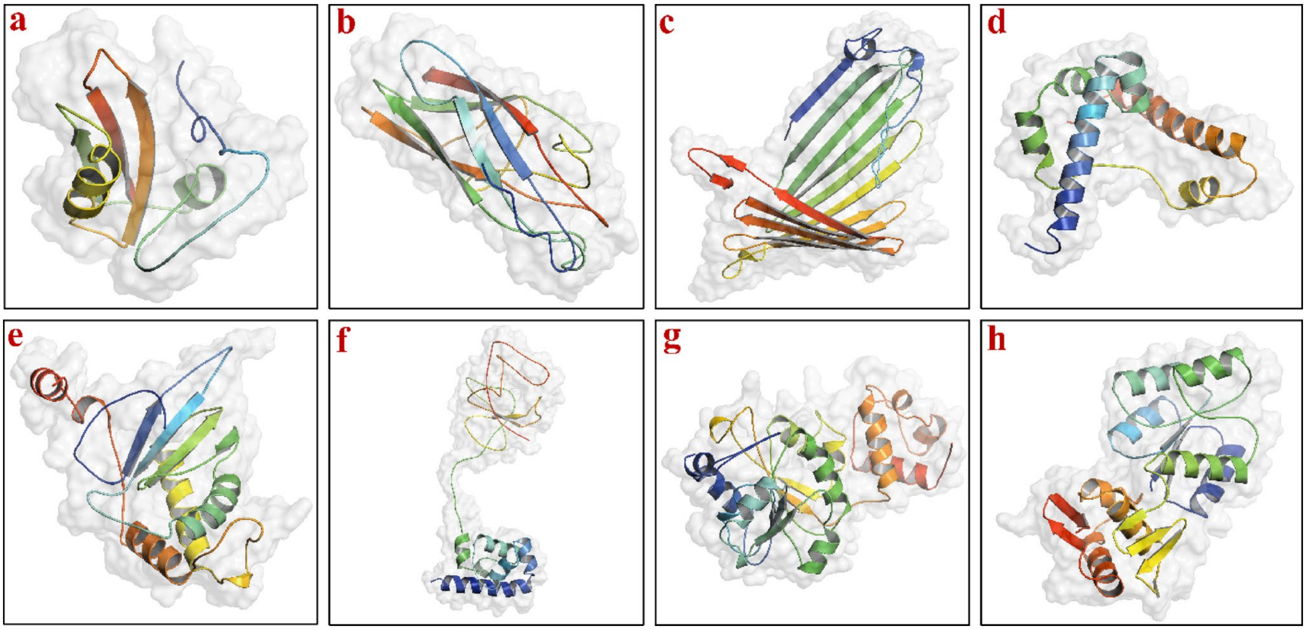

Fig. 1 Predicted three-dimensional structures of quorum-sensing proteins modeled using I-TASSER: **a** BsmA, **b** FimA, **c** FimC, **d** FlhD, **e** LuxS, **f** RsmA, **g** PigP, and **h** RssB

Table 3 Docking score of proteins with coumaric and syringic acid

Protein	Coumaric acid	Syringic acid
bsmA	−4.9	−4.7
fimA	−4.7	−4.8
fimC	−5.6	−5.3
flhD	−5.5	−5.2
luxS	−5.7	−5.0
pigP	−5.8	−4.9
rsmA	−6.1	−5.9
rssB	−5.5	−5.0

of 4.3. The physicochemical parameters of both ligands are summarized in Supplementary Table 1. Molecular docking, performed using AutoDock Vina within the UCSF Chimera environment, generated five conformations for each ligand–protein complex, from which the lowest-energy conformation was selected for molecular dynamics (MD) simulations (Table 3). Representative surface views of the protein–ligand complexes are shown in Fig. 2. Key molecular interactions within the selected complexes were further analysed (Table 4), with interaction maps provided in Supplementary Fig. 2. Additionally, 2D interaction maps were created

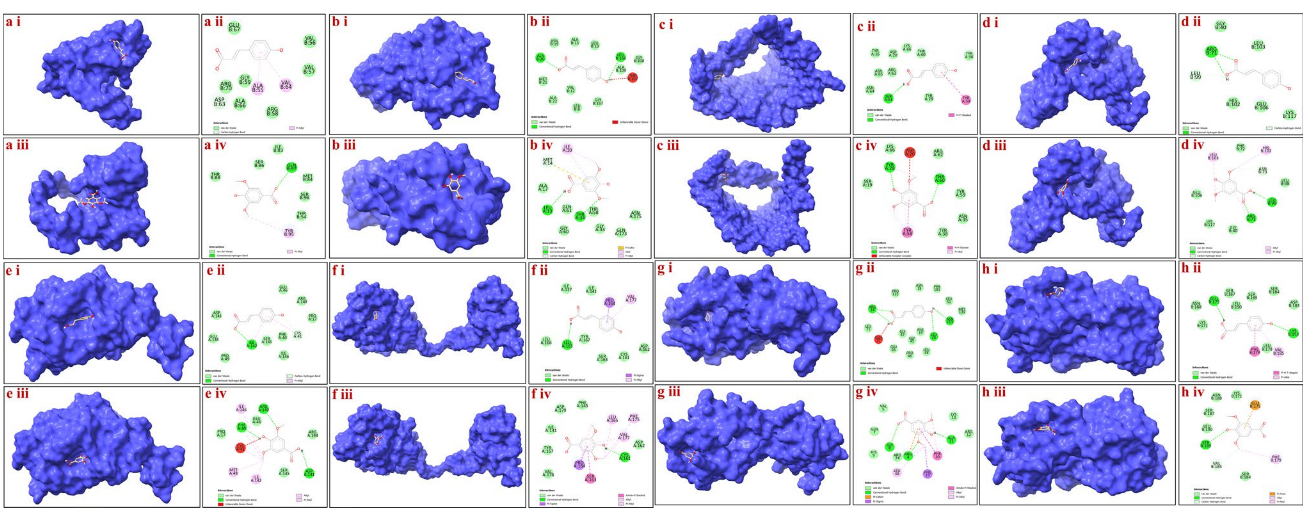

Fig. 2 Surface representations and 2D interaction maps of protein–ligand complexes. For each protein, panels (i) and (iii) show the surface structure with coumaric acid and syringic acid bound in the predicted binding pocket, respectively. Panels (ii) and (iv) display the corresponding 2D interaction maps, illustrating hydrogen bonds, hydrophobic interactions, and key residues involved in ligand binding. Proteins analyzed include: **a** BsmA, **b** FimA, **c** FimC, **d** FlhD, **e** LuxS, **f** PigP, **g** RsmA, and **h** RssB

Table 4 Type of interactions between proteins and coumaric acid and Syringic acid

Proteins	Coumaric acid		Syringic acid	
	Hydrogen bond	Hydrophobic interaction	Hydrogen bond	Hydro-phobic interaction
bsmA	1	3	1	1
fimA	5	1	3	–
fimC	2	3	3	2
flhD	2	2	2	3
luxS	1	2	2	3
pigP	2	5	3	3
rsmA	3	3	1	2
rssB	1	4	2	2

using BIOVIA Discovery Studio to visually represent these interactions (also shown in Fig. 2).

3.2 Molecular Dynamic Simulations

MD simulations (100 ns) were performed for all protein–ligand complexes to assess the binding behaviour of coumaric and syringic acids. Trajectories were analysed for RMSD, RMSF, Rg, SASA, DSSP, HB, MM/GBSA, and PCA.

3.3 Root Mean Square Deviation

RMSD analysis was used to evaluate the effect of ligand binding on protein stability during MD simulations (Fig. 3a). Binding of coumaric acid generally led to elevated RMSD

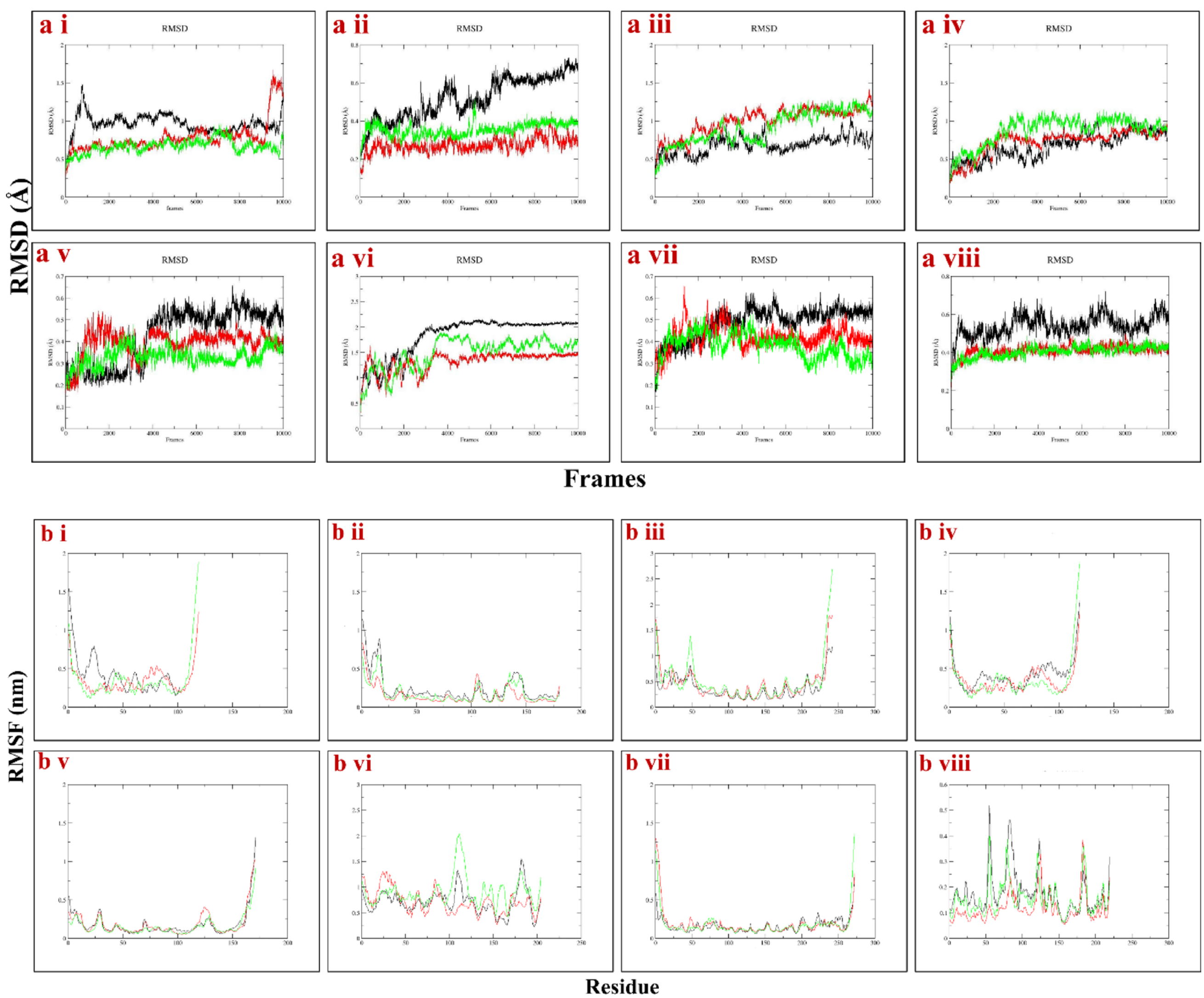

Fig. 3 Molecular dynamics trajectory analyses of protein–ligand complexes. **a** Root Mean Square Deviation (RMSD) plots for proteins bound to coumaric acid and syringic acid: (i) BsmA, (ii) FimA, (iii) FimC, (iv) FlhD, (v) LuxS, (vi) PigP, (vii) RsmA, and (viii) RssB. **b** Root Mean Square Fluctuation (RMSF) plots for the same protein–ligand complexes: (i) BsmA, (ii) FimA, (iii) FimC, (iv) FlhD, (v) LuxS, (vi) PigP, (vii) RsmA, and (viii) RssB. In all graphs, black lines represent coumaric acid–protein complexes, green lines represent syringic acid–protein complexes, and red lines represent unbound (apo) proteins

values in six proteins—BsmA (0.964 ± 0.123 Å; Fig. 3a-i), FimA (0.518 ± 0.110 Å; Fig. 3a-ii), LuxS (0.426 ± 0.128 Å; Fig. 3a-v), PigP (1.823 ± 0.389 Å; Fig. 3a-vi), RsmA (0.486 ± 0.073 Å; Fig. 3a-vii), and RssB (0.540 ± 0.056 Å; Fig. 3a-viii)—indicating greater conformational fluctuations compared to syringic acid or apo forms. In contrast, FimC (0.915 ± 0.219 Å; Fig. 3a-iii) and FlhD (0.876 ± 0.184 Å; Fig. 3a-iv) exhibited higher RMSD values when bound to syringic acid, reflecting ligand-specific effects on structural dynamics. Notably, apo FimC (0.994 ± 0.184 Å; Fig. 3a-iii) showed marked instability in the absence of ligand, suggesting inherent structural drift. Among the ligand-bound systems, the BsmA–coumaric acid and PigP–coumaric acid complexes displayed the highest RMSD values (Table 5), strongly supporting the notion that coumaric acid induces substantial conformational changes. Overall, coumaric acid destabilized six of the eight proteins studied, whereas syringic acid induced instability in FimC and FlhD.

3.4 Root Square Mean Fluctuations (RMSF)

RMSF analysis was performed to assess residue-level flexibility in the protein–ligand complexes (Fig. 3b). Lower RMSF values indicate residue rigidity and stabilization through ligand binding, whereas higher values reflect enhanced flexibility and reduced stabilization. Coumaric acid binding resulted in reduced per-residue fluctuations in FimC (Fig. 3b-iii), PigP (Fig. 3b-vi), and RsmA (Fig. 3b-vii) compared with both apo forms and syringic acid complexes, indicating enhanced structural rigidity. In contrast, BsmA (Fig. 3b-i), FimA (Fig. 3b-ii), FlhD (Fig. 3b-iv), LuxS (Fig. 3b-v), and RssB (Fig. 3b-viii) exhibited greater stabilization in the presence of syringic acid. Overall, coumaric acid tend to enhance global stability, whereas syringic acid generally induced localized flexibility. Notably, coumaric acid increased flexibility at the N-terminal regions, while syringic acid binding promoted C-terminal destabilization in most proteins (Fig. 3b). Average RMSF values are summarized in Table 5.

3.5 Radius of Gyration (Rg)

The radius of gyration (Rg) was analysed to assess protein compactness, with lower Rg values indicating more compact structures (Fig. 4a). Across the systems, ligand binding induced measurable conformational changes relative to the apo proteins. Consistently higher Rg values were observed in syringic acid complexes of BsmA (1.539 ± 0.059 Å; Fig. 4a-i), FimC (2.401 ± 0.158 Å; Fig. 4a-iii), FlhD (1.745 ± 0.088 Å; Fig. 4a-iv), LuxS (1.754 ± 0.025 Å; Fig. 4a-v), PigP (2.586 ± 0.352 Å; Fig. 4a-vi), and RsmA (2.122 ± 0.027 Å; Fig. 4a-vii), suggesting a tendency toward more open or flexible conformations. In contrast, coumaric acid complexes generally exhibited lower Rg values than their syringic acid counterparts, with the exceptions of FimA (1.765 ± 0.032 Å; Fig. 4a-ii) and RssB (1.939 ± 0.016 Å;

Table 5 Average values of MD trajectories of RMSD, RMSF, Rg and SASA

Protein/protein–ligand	RMSD (Å)	RMSF (nm)	Rg (nm)	SASA (nm^2)
bsmA	0.780 ± 0.203	0.458 ± 0.245	1.486 ± 0.151	79.222 ± 4.454
bsmA-Coumaric acid	0.964 ± 0.123	0.418 ± 0.242	1.537 ± 0.091	77.991 ± 4.344
bsmA-Syringic acid	0.668 ± 0.088	0.362 ± 0.199	1.539 ± 0.059	79.322 ± 2.887
fimA	0.274 ± 0.038	0.154 ± 0.121	1.744 ± 0.015	102.039 ± 2.238
fimA-Coumaric acid	0.518 ± 0.110	0.236 ± 0.190	1.765 ± 0.032	105.130 ± 3.884
fimA-Syringic acid	0.355 ± 0.035	0.178 ± 0.129	1.752 ± 0.034	108.041 ± 3.178
fimC	0.994 ± 0.184	0.437 ± 0.318	2.122 ± 0.131	169.175 ± 9.048
fimC-Coumaric acid	0.687 ± 0.111	0.372 ± 0.220	2.324 ± 0.092	179.493 ± 5.104
fimC-Syringic acid	0.915 ± 0.219	0.498 ± 0.413	2.401 ± 0.158	183.713 ± 9.001
flhD	0.726 ± 0.163	0.350 ± 0.174	1.658 ± 0.090	92.489 ± 6.009
flhD-Coumaric acid	0.671 ± 0.172	0.426 ± 0.203	1.730 ± 0.056	94.987 ± 5.478
flhD-Syringic acid	0.876 ± 0.184	0.361 ± 0.299	1.745 ± 0.088	93.537 ± 3.429
luxS	0.393 ± 0.061	0.185 ± 0.167	1.740 ± 0.041	103.968 ± 2.370
luxS-Coumaric acid	0.426 ± 0.128	0.179 ± 0.170	1.728 ± 0.024	101.984 ± 2.522
luxS-Syringic acid	0.330 ± 0.042	0.158 ± 0.122	1.754 ± 0.025	103.968 ± 2.370
pigP	1.324 ± 0.184	0.705 ± 0.232	2.355 ± 0.254	141.182 ± 3.409
pigP-Coumaric acid	1.823 ± 0.389	0.705 ± 0.235	2.034 ± 0.251	142.433 ± 9.249
pigP-Syringic acid	1.495 ± 0.306	0.895 ± 0.286	2.586 ± 0.352	160.696 ± 5.971
rsmA	0.420 ± 0.054	0.182 ± 0.166	2.112 ± 0.037	147.855 ± 4.520
rsmA-Coumaric acid	0.486 ± 0.073	0.162 ± 0.088	2.096 ± 0.016	145.249 ± 3.837
rsmA-Syringic acid	0.380 ± 0.058	0.174 ± 0.169	2.122 ± 0.027	147.023 ± 3.979
rssB	0.410 ± 0.029	0.117 ± 0.055	1.854 ± 0.017	123.328 ± 3.249
rssB-Coumaric acid	0.540 ± 0.056	0.174 ± 0.080	1.939 ± 0.016	133.330 ± 3.198
rssB-Syringic acid	0.404 ± 0.031	0.153 ± 0.065	1.864 ± 0.019	126.390 ± 3.636

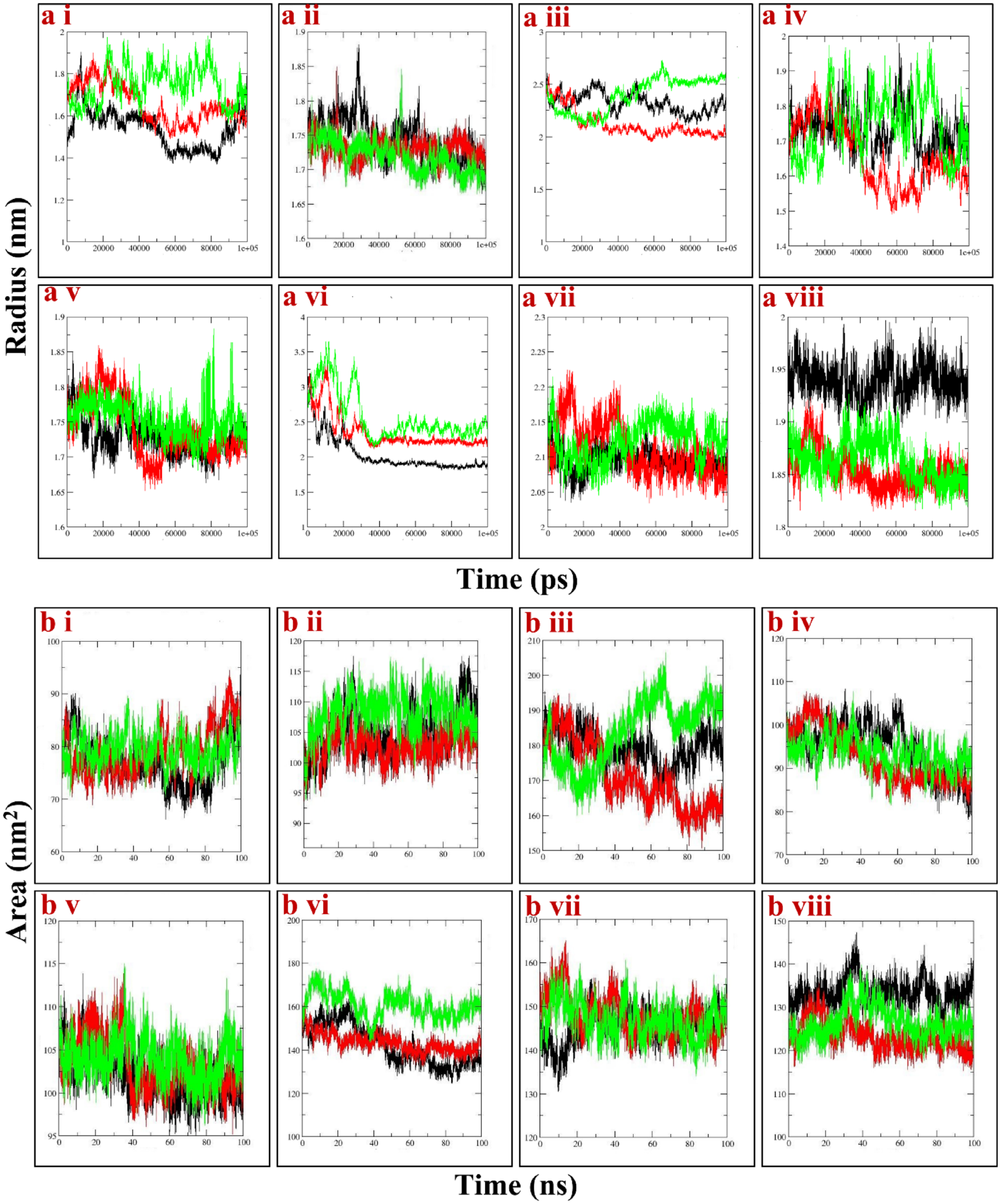

Fig. 4 Molecular dynamics trajectory analyses of protein–ligand complexes in terms of structural compactness and solvent exposure. **a** Radius of Gyration (Rg) plots for proteins bound to coumaric acid and syringic acid: (i) BsmA, (ii) FimA, (iii) FimC, (iv) FlhD, (v) LuxS, (vi) PigP, (vii) RsmA, and (viii) RssB. **b** Solvent Accessible Surface Area (SASA) plots for the same complexes: (i) BsmA, (ii) FimA, (iii) FimC, (iv) FlhD, (v) LuxS, (vi) PigP, (vii) RsmA, and (viii) RssB. In all graphs, black lines represent coumaric acid–protein complexes, green lines represent syringic acid–protein complexes, and red lines represent unbound (apo) proteins

Fig. 4a-viii), indicating greater structural compactness. Temporal fluctuations in Rg further demonstrated that all systems sampled multiple conformations during the simulations, with larger fluctuations associated with reduced structural stability. Average Rg values for all proteins are summarized in Table 5.

3.6 Solvent Accessible Surface Area (SASA)

SASA analysis was performed to evaluate the solvent-exposed surface area of the proteins, with changes in SASA reflecting conformational rearrangements such as folding, unfolding, or ligand binding (Fig. 4b). Fluctuations observed across all systems indicated dynamic behavior throughout the simulations. Elevated SASA values were consistently observed in the syringic acid complexes of BsmA (79.322 ± 2.887 $Å^2$; Fig. 4b-i), FimA (108.041 ± 3.178 $Å^2$; Fig. 4b-ii), FimC (183.713 ± 9.001 $Å^2$; Fig. 4b-iii), LuxS (103.968 ± 2.370 $Å^2$; Fig. 4b-iv), PigP (160.696 ± 5.971 $Å^2$; Fig. 4b-vi), and RsmA (147.023 ± 3.979 $Å^2$; Fig. 4b-vii), suggesting that syringic acid binding promotes a more extended conformation and greater solvent accessibility. In contrast, coumaric acid binding generally reduced SASA in these proteins, consistent with a more compact conformation. Exceptions were observed in FlhD–coumaric acid (93.537 ± 3.429 $Å^2$; Fig. 4b-iv) and RssB–coumaric acid (126.390 ± 3.636 $Å^2$; Fig. 4b-viii), where increased SASA indicated ligand-induced structural expansion. Overall, higher SASA values reflect enhanced hydrophilic exposure, whereas reduced SASA suggests increased hydrophobic packing. Average SASA values are summarized in Table 5.

3.7 Hydrogen Bonding

Hydrogen bond analysis demonstrated that coumaric acid formed consistent and abundant hydrogen bonds with seven proteins (Supplementary Fig. 3a-i–vii), whereas syringic acid established a stable hydrogen bond only with RssB (Supplementary Fig. 3b-viii). The persistence of these interactions throughout the simulation highlights the stable incorporation of coumaric acid within the binding pockets and suggests stronger binding affinity relative to syringic acid (Supplementary Fig. 3).

3.8 Protein–Phenolic Compounds MMGBSA Binding Energy

Binding free energy calculations were performed to assess ligand–protein interactions, with negative ΔG_bind values indicating favorable binding (Fig. 5). Coumaric acid exhibited higher binding affinity toward six proteins—BsmA, FimA, FimC, LuxS, PigP, and RsmA—whereas syringic acid showed greater affinity for RssB. FlhD displayed nearly equivalent affinities for both ligands. The LuxS–coumaric acid complex showed the strongest binding (−21.74 ± 3.01 kcal/mol), while the FimA–syringic acid complex displayed the weakest (−2.67 ± 4.58 kcal/mol). These results suggest that coumaric acid generally achieves stronger and more stable interactions, likely due to enhanced complementarity within the binding pocket. ΔG_bind profiles for individual proteins are shown in Supplementary Fig. 4a-i–viii (coumaric acid) and Fig. 4b-i–viii (syringic acid), with mean values summarized in Supplementary Table 2.

3.9 Dictionary of Secondary Structure of Proteins (DSSP)

DSSP analysis was conducted to evaluate the impact of ligand binding on protein secondary structure (Supplementary Fig. 5). Coumaric acid consistently stabilized α-helices and β-sheets across proteins, whereas syringic acid induced notable coil/turn transitions, particularly in FlhD (Supplementary Fig. 5d) and LuxS (Supplementary Fig. 5e), indicating disruption of ordered motifs. Apo proteins displayed intermediate behaviour with partial destabilization in loop-rich regions. Overall, coumaric acid binding better preserved secondary structure integrity compared to syringic acid.

3.10 Principal Component Analysis

Principal component analysis (PCA) reduces high-dimensional atomic fluctuations into principal components (PCs), enabling visualization of conformational space sampling. Compact clustering of PCs reflects restricted motions and structural stability, whereas dispersed distributions indicate enhanced flexibility and reduced stability. PCA revealed marked differences in the conformational dynamics of protein complexes with coumaric acid and syringic acid (Supplementary Fig. 4). In BsmA (Fig. 4a-i), coumaric acid displayed a broad trajectory, reflecting multiple stable states, whereas syringic acid exhibited a compact but less stable profile. FimA (Fig. 4a-ii) and RsmA (Fig. 4a-vii) showed well-defined clusters with coumaric acid, indicative of strong stabilization, while syringic acid produced scattered, less defined conformations. For FimC (Fig. 4a-iii) and FlhD (Fig. 4a-iv), coumaric acid facilitated transitions among stable conformational basins, whereas syringic acid yielded diffuse and unstable distributions. Similarly, LuxS (Fig. 4a-v) and PigP (Fig. 4a-vi) complexes with coumaric acid exhibited restricted sampling and discrete states, contrasting with the broad scattering observed for syringic acid. In RssB (Fig. 4a-viii), coumaric acid maintained compact

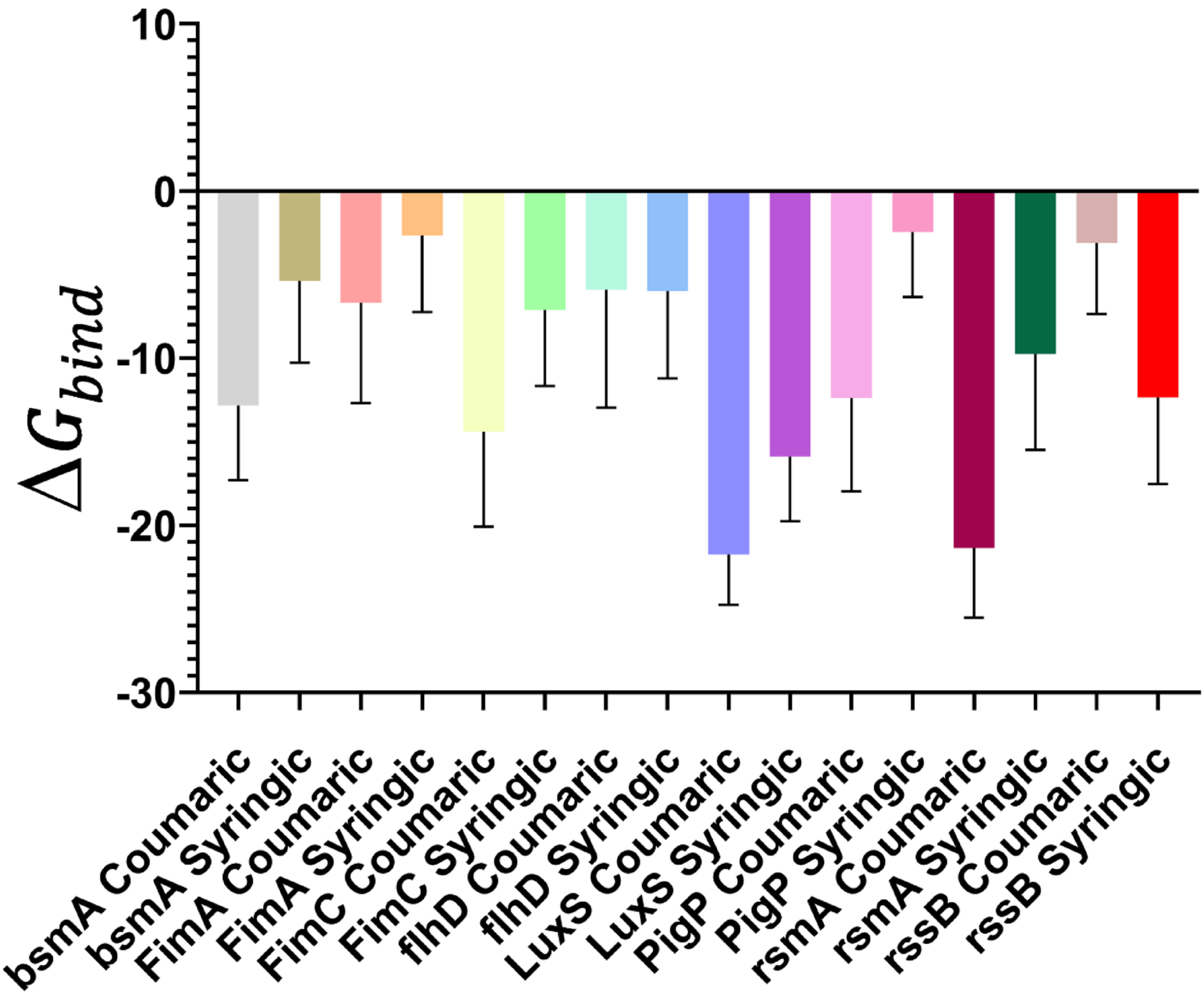

Fig. 5 Mean ΔG_{bind} graph for protein-coumaric and protein-syringic acid complex. (protein–ligand complex, denoted by the black line, whereas the red line signifies the average value)

conformational basins, while syringic acid induced instability. Collectively, these results demonstrate that coumaric acid consistently stabilizes protein structures by promoting well-clustered conformational states, whereas syringic acid tends to increase conformational heterogeneity and instability.

3.11 Antibacterial and anti-quorum sensing potential of phenolic acids in S. marcescens

T he minimum inhibitory concentration (MIC) of coumaric acid and syringic acid against the respiratory tract pathogen *Serratia marcescens* was determined as 700 µg/mL and 1000 µg/mL, respectively. These results demonstrate that coumaric acid exhibits greater antimicrobial potency, as it inhibits visible bacterial growth at a lower concentration compared to syringic acid.

The minimum bactericidal concentration (MBC) assay was performed to differentiate between bactericidal and bacteriostatic effects of these compounds. Coumaric acid exhibited bactericidal activity, completely eliminating viable *S. marcescens* cells, whereas syringic acid was bacteriostatic, suppressing bacterial growth without inducing cell death. These findings emphasize coumaric acid as a more potent candidate for therapeutic intervention against *S. marcescens*- associated respiratory infections.

3.12 Growth Kill Assay

The antibacterial effects of coumaric acid and syringic acid was further assessed using a growth-kill assay. Absorbance-time curves demonstrated a concentration-dependent inhibition of bacterial growth over 24-h. Initially, all treated samples showed similar growth kinetics; however, progressive reductions in viable cell numbers were observed with increasing compound concentrations. Notably, coumaric acid completely suppressed bacterial growth at concentrations ≥ 500 µg/mL. In comparison, syringic acid required > 1000 µg/mL to achieve similar inhibitory effect (Fig. 6).

3.13 Phenolic compounds attenuate S. Marcescens pathogenicity by down-regulation of QS-associated gene expression

To explore the anti-QS potential of coumaric acid and syringic acid, quantitative real-time PCR (qRT-PCR) was performed to measure the expression of key QS-regulated genes in *Serratia marcescens*. Transcript levels of luxS, pigP, fimA, fimC, rsmA, flhD, and bsmA,- involved in QS-mediated biofilm formation, motility, and virulence, were quantified following treatment with coumaric acid and syringic acid at sub-inhibitory concentrations (1/8th, 1/4th, and 1/2 MIC). Both compounds exhibited significant anti-QS activity by downregulating QS-controlled genes.Notably, luxS, a key gene for QS autoinducer synthesis, showed a 6-fold and 8-fold reduction in expression upon treatment with coumaric acid and syringic acid at 1/2 MIC, respectively. The pigment biosynthesis regulator pigP, which is QS-dependent, was downregulated by 6.4-fold with coumaric acid and 20-fold with syringic acid, indicating a strong inhibition of QS-controlled pigment production (Fig. 7). Furthermore, genes involved in QS-mediated biofilm development and structural integrity (bsmA, fimA, and fimC) were significantly suppressed. bsmA was reduced by 5.73-fold with coumaric acid and 16-fold with syringic acid, suggesting impaired biofilm maturation. Similarly, the fimbrial adhesion genes fimA and fimC were downregulated by 2-fold and 12.64-fold with coumaric acid, and by 22-fold and 8-fold with syringic acid, respectively, indicating reduced bacterial attachment and surface colonization. Additionally, rsmA, a regulator of motile-to-sessile transition, exhibited a 2.36-fold reduction with coumaric acid and a 22-fold reduction with syringic acid, while flhD, essential for flagellar motility, was downregulated by 3.77-fold and 16-fold, respectively. Collectively, these results demonstrate that both coumaric acid and syringic acid strongly inhibit QS-regulated gene networks, thereby attenuating biofilm formation, motility, and virulence in *S. marcescens* (Fig. 7).

4 Discussion

Antibiotic resistance (AR) poses a grave global threat, diminishing the efficacy of antibiotics and complicating disease treatment. With over 2.8 million infections annually in the United States alone, AR leads to more than 35,000 deaths. The economic toll is significant, with projections suggesting a potential $100 trillion global cost by 2050 [58]. Extended exposure to sublethal concentrations of antibiotics, driven by irrational antibiotic usage, is a key factor in bacterial resistance selection [59, 60]. In 2021, an estimated 4.71 million deaths were associated with bacterial AMR, of which 1.14 million were directly attributable to AMR alone. On an annual scale, AMR is responsible for approximately 4.95 million deaths worldwide [61]. The most resilient pathogens driving this global crisis remain the ESKAPE group—*Enterococcus faecium*, *Staphylococcus aureus*, *Klebsiella pneumoniae*, *Acinetobacter baumannii*, *Pseudomonas aeruginosa*, and *Enterobacter spp.*—notorious for their ability to evade multiple antibiotics and cause severe nosocomial infections [62]. The rapid rise of these drug-resistant strains underscores the urgent need for new treatment strategies and stronger efforts to preserve the effectiveness of existing antibiotics. *Serratia marcescens*, a notorious ESKAPE pathogen, poses significant challenges

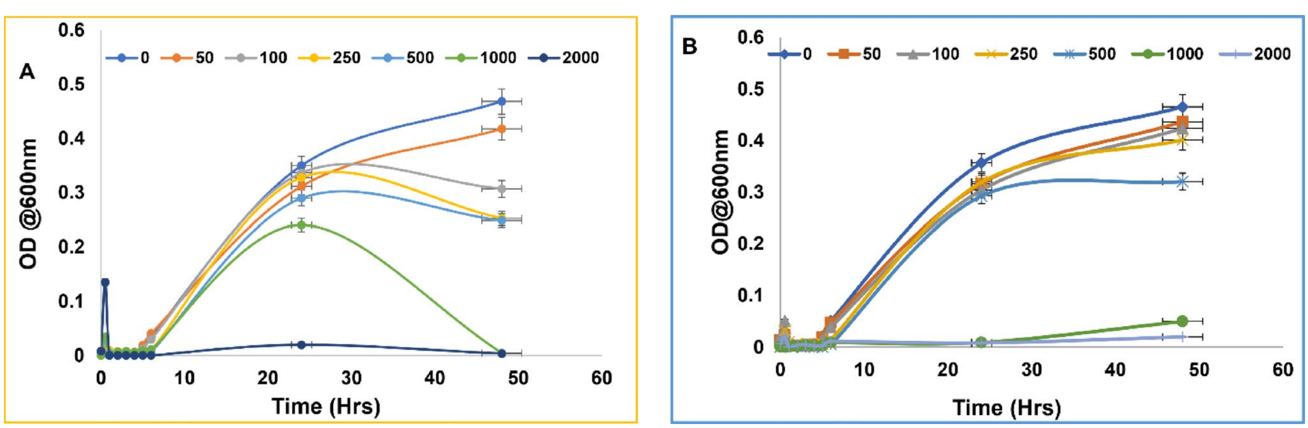

Fig. 6 Time-kill curves illustrating the impact of phenolic acids on the growth kinetics of *Serratia marcescens*: **a** Time-dependent antibacterial effect of coumaric acid; **b** Time-dependent antibacterial effect of syringic acid. Data represent the mean of three independent experiments, and statistical significance was determined at $P < 0.05$

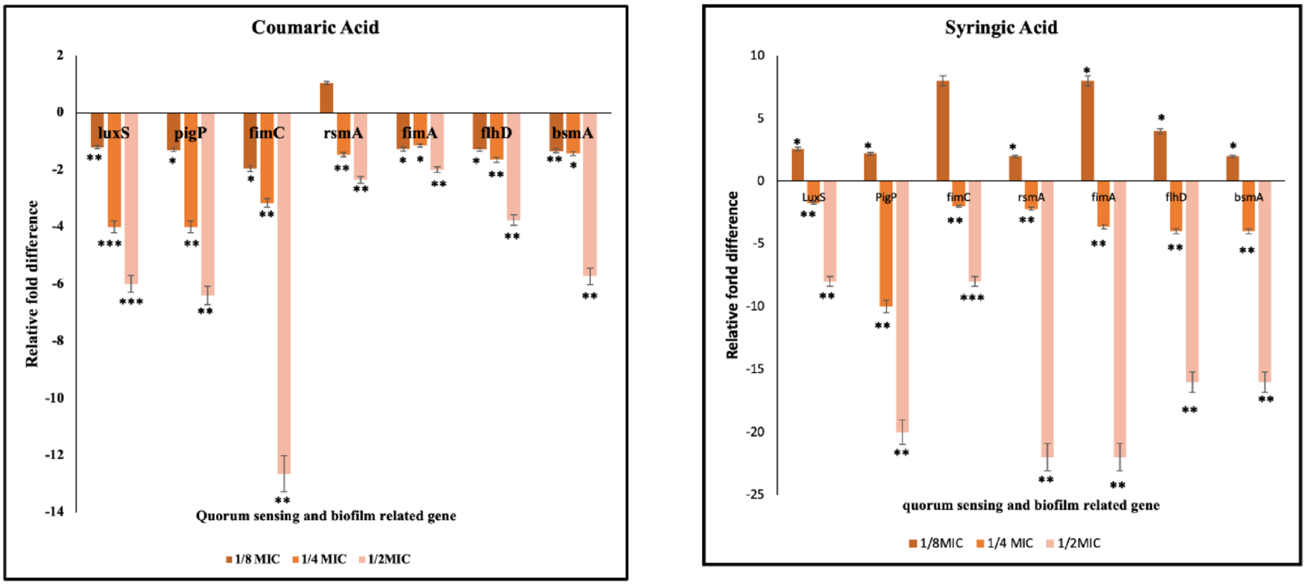

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Fig. 7 Quantitative real-time PCR analysis reveals altered expression of quorum sensing and biofilm-associated genes in *Serratia marcescens* following phenolic acid treatment. **a** Relative mRNA expression levels of QS and biofilm regulon genes after coumaric acid treatment; **b** Relative mRNA expression levels of QS and biofilm regulon genes after syringic acid treatment. Data represent the mean of three independent experiments, and statistical significance was determined at 12; $^{*}P<0.05$; $^{**}P<0.01$; and $^{***}P<0.001$

due to its ability to cause nosocomial infections and develop resistance to multiple antibiotics. Its widespread presence in healthcare settings contributes to difficult-to-treat infections, especially in immunocompromised individuals. Its ability to form robust biofilms enhances resilience and hampers treatment efficacy, underscoring the urgent need for targeted interventions [63].

Plant-derived compounds present a rich array of chemical structures that exhibit potent antimicrobial activity against resistant bacteria. Their multifaceted mechanisms, targeting biofilm disruption, growth inhibition, and QS-induced virulence factor suppression, make them promising candidates for novel antimicrobial agents. Notably, their efficacy comes with minimal side effects and enduring impact, offering sustainable solutions to the growing challenge of antimicrobial resistance [64–69]. The present study investigates the impact of coumaric acid and syringic acid on eight quorum-sensing (QS) proteins of Serratia marcescens—BsmA, FimA, FimC, FlhD, LuxS, PigP, RsmA, and RssB—using a combination of in silico molecular dynamics simulations and in vitro antibacterial assays. Homology modelling, molecular docking, and MD simulations were employed to comparatively evaluate the binding characteristics of these two phenolic acids across the protein panel.

ADME profiling and docking analyses revealed distinct binding characteristics of the two ligands. Coumaric acid demonstrated favourable pharmacokinetic attributes—optimal lipophilicity (logP 1.79), moderate polarity (TPSA 57.53 Å^2), and suitable ionization (pKa 4.64)—which were consistent with its lower binding energies across seven proteins (Table 3) and stronger interaction stability. In contrast, syringic acid, although predicted to have higher absorption, displayed reduced bioavailability (logP 1.04; TPSA 75.99 Å^2), correlating with weaker docking scores and less stable protein–ligand complexes (Supplementary Table 1). To further evaluate these interactions, 100-ns MD simulations were performed, providing detailed insights into the dynamic stability of the complexes.

RMSD analysis revealed that coumaric acid induced more pronounced conformational rearrangements in six proteins, most notably BsmA and PigP (Fig. 2 a i and 2 a vi) as evidenced by the highest RMSD values (Table 4). In contrast, syringic acid complexes with FimC and FlhD displayed elevated RMSD (Fig. 2 a iii and 2 a iv). While such increases might suggest destabilization, complementary RMSF, Rg, and DSSP analyses indicate that these deviations largely represent ligand-induced conformational adaptations (induced fit) rather than uncontrolled structural drift (Supplementary Fig. 5). Singh et al. [68] reported enhanced stability of coumaric acid binding to the NorA protein, with an RMSD of 4.2 Å, while Akpinar et al. [67] demonstrated its antibacterial activity through stable interactions with DNA gyrase subunit B, characterized by RMSD values below 2 Å. In comparison, the present study revealed RMSD values below 1 Å for nearly all coumaric acid–protein complexes, except PigP, indicating an even greater degree of structural stability than previously reported.

RMSF analysis demonstrated that coumaric acid reduced residue-level fluctuations in most proteins, thereby enhancing structural rigidity and stability. By contrast, syringic acid frequently promoted localized flexibility, particularly within C-terminal regions, which may compromise structural stability and function (Fig. 2b). Supporting these findings, Rg and SASA analyses showed that coumaric acid binding generally produced more compact and hydrophobic conformations, whereas syringic acid complexes displayed expanded structures with higher solvent exposure, indicative of reduced packing efficiency (Table 5). Previous studies on plant-derived compounds reported RMSF values in the range of 3–7 Å for coumaric acid–NorA complexes in various Gram-positive and Gram-negative bacteria [70]. In comparison, our analysis showed RMSF values ranging from 1–10 Å for coumaric acid–docked structures and 1–12 Å for syringic acid complexes (Fig. 3b), indicating greater fluctuation than previously reported. Collectively, these observations suggest that coumaric acid promotes tighter structural organization conducive to stable protein–ligand interactions, while syringic acid tends to induce conformational looseness and decreased stability.

Hydrogen bond profiling revealed that coumaric acid consistently formed persistent interactions across seven protein complexes, emphasizing its role in sustaining structural stability during MD simulations. In comparison, syringic acid exhibited stable hydrogen bonding only with RssB, indicative of a markedly reduced stabilizing effect. (Supplementary Fig. 3). Binding free energy provides critical insights into protein–ligand interactions, with negative values denoting favourable binding and lower ΔG values reflecting stronger affinities [71, 72]. MM/GBSA binding free energy calculations reinforced the preferential binding of coumaric acid, which exhibited more negative ΔG_bind values across six proteins, with the LuxS–coumaric acid complex showing the strongest affinity (-21.74 ± 3.01 kcal/mol) (Supplementary Table 2). Syringic acid displayed stronger binding only with RssB, while FlhD showed comparable affinities for both ligands (Fig. 5). These findings suggest that coumaric acid more effectively occupies binding pockets, thereby enhancing complex stability and inhibitory potential. Structural analyses provided converging evidence for this trend. DSSP revealed that coumaric acid consistently preserved α-helices and β-sheets, maintaining key structural motifs, whereas syringic acid promoted coil/turn transitions in FlhD and LuxS, indicative of destabilization (Supplementary Fig. 4). PCA further highlighted these differences at the level of conformational dynamics: coumaric acid complexes113 formed compact, well-defined conformational basins, reflecting restricted sampling and long-term stability, while syringic acid complexes displayed diffuse trajectories consistent with increased flexibility and reduced stability (Supplementary Fig. 6 a, 6 b). Notably, coumaric acid induced distinct, stable conformational states in FimA, LuxS, and RsmA—proteins critical for *S. marcescens* virulence—underscoring its potential as an effective antibacterial agent.

In vitro biological assays validated the computational predictions and revealed distinct antimicrobial mechanisms of coumaric and syringic acids in *S. marcescens*. MIC and MBC assays confirmed the bactericidal nature of coumaric acid, while syringic acid exhibited only bacteriostatic activity. Growth–kill assays further demonstrated that coumaric acid completely suppressed bacterial growth at concentrations ≥ 500 µg/mL, whereas syringic acid required ≥ 1000 µg/mL to achieve comparable inhibition, confirming the superior killing efficiency of coumaric acid (Fig. 6). Remarkably, despite its weaker antibacterial effect, syringic acid exerted significantly stronger transcriptional suppression of QS-related genes. qRT-PCR analysis showed dramatic downregulation of pigP (−20-fold), fimA (−22-fold), rsmA (−22-fold), flhD (−16-fold), and bsmA (−16-fold), far exceeding the moderate effects of coumaric acid (luxS −6-fold; pigP −6.4-fold; bsmA −5.7-fold) (Fig. 7). These findings reveal a clear functional divergence: coumaric acid primarily acts as a bactericidal agent by destabilizing regulatory proteins and promoting cell death, whereas syringic acid functions as a potent quorum-quenching agent, attenuating virulence factors such as pigment production, motility, and biofilm formation without directly killing the cells.

These outcomes are consistent with accumulating evidence on the quorum-quenching potential of plant-derived compounds. Flavonoids such as flavone, hesperidin methyl chalcone, morin, and quercetin have been shown through molecular dynamics simulations to interfere with QS pathways, reducing virulence and biofilm formation in pathogens including *Pseudomonas aeruginosa*, *Aeromonas hydrophila*, and *Staphylococcus aureus* [73–77]. Large-scale computational screenings have further identified diverse flavonoids with activity against drug-resistant bacteria., while other phytochemicals such as quercetin, ferulic acid, and gallic acid have also been shown to suppress QS-regulated virulence in *P. aeruginosa* and *S. aureus*. Collectively, these findings reinforce the anti-QS and anti-biofilm potential of phytochemicals as an emerging strategy for antimicrobial development [78–88]. Notably, Kim et al. reported that gingerol, a bioactive component of ginger, binds to LasR and competitively inhibits its activation, leading to reduced virulence factor production and biofilm formation [89].

Taken together with this accumulating evidence, our results advance the understanding of natural compounds as therapeutic candidates, demonstrating that coumaric acid functions as a potent antimicrobial while syringic acid operates as a targeted anti virulence agent. This duality underscores the therapeutic promise of plant-derived molecules at the interface of plant chemistry and microbiology, opening innovative avenues for combating persistent bacterial infections.

5 Conclusion

This study demonstrates the stable interactions of coumaric acid and syringic acid with eight quorum-sensing proteins of *Serratia marcescens* (BsmA, FimA, FimC, FlhD, LuxS, PigP, RsmA, and RssB) through computational analyses. Integrating computational and experimental data, coumaric acid is established as the superior antibacterial candidate, characterized by lower MIC values, confirmed bactericidal activity, strong binding affinity, persistent hydrogen bonding, and pronounced stabilizing effects on protein structures. In contrast, syringic acid exhibits weaker antibacterial potency but functions as a potent quorum-quenching agent, markedly downregulating QS-regulated genes that control virulence, motility, pigment production, and biofilm formation.

Taken together, these findings position coumaric acid as a promising lead for direct antibacterial therapy, while syringic acid represents a valuable anti virulence adjunct capable of suppressing pathogenicity without imposing bactericidal pressure that could accelerate resistance. A synergistic therapeutic strategy combining coumaric acid's bactericidal action with syringic acid's quorum-quenching activity may provide an innovative approach to combat multidrug-resistant *S. marcescens* infections. However, the present work is limited to *in silico* and in vitro analyses; further in vivo validation, toxicity profiling, and mechanistic studies are required to confirm the therapeutic potential of these compounds.

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Data Availability The data can be accessed upon reasonable request to the corresponding author.

Declarations

Conflict of interest The authors confirm that there are no conflicts of interest in the publication of this work.

Ethical Approval This study did not involve any experiments on human participants or animals conducted by the authors.

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